Breathless but Not Blind: Interpreting Multi-Signal Patterns of Sleep Apnea with a Sensor-Robust Peripheral-Central Spiking Network

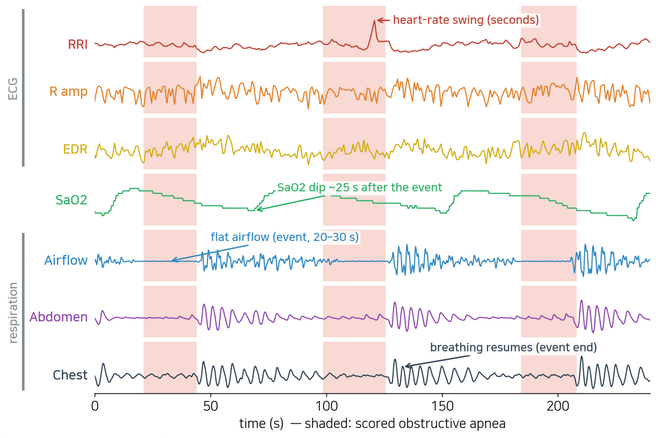
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HAI Lab

1. INTRODUCTION

Problem

Sleep apnea affects ~1 billion adults, mostly undiagnosed. A PSG(Polysomnography) night shows the same event with different time signatures in each signal.



Limitations of current models

(1) One time scale for all signals
(2) Black-box fusion: No signal attribution
(3) Fails when SaO2 is missing

This work

Peripheral-central spiking network that:
(1) Per-signal streams on own time scale
(2) Structure shows signal combinations
(3) STF gate → works without SaO2

2. DATASET

PhysioNet / CinC Challenge 2018

- 994 annotated overnight PSGs · Massachusetts General Hospital (MGH) sleep laboratory,

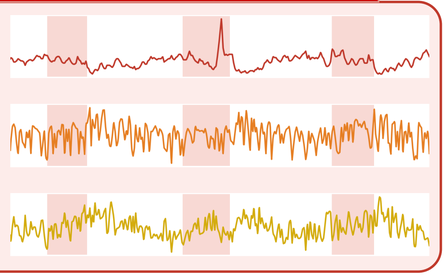
Dataset: 994

Normal : 83.9% | Apena: 16.1%

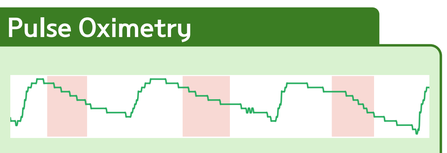
Training: 793 | Test: 201

Training: 694 | Valid: 99 | Test: 201

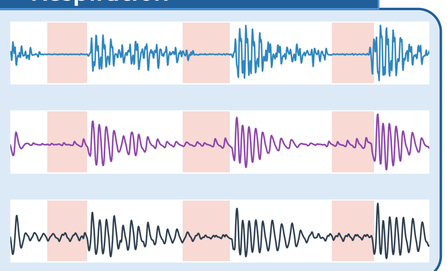
ECG (single lead) → 3 signals



Pulse Oximetry



Respiration



Label

A 1-min unit is apnea if a scored apnea or hypo-apnea overlaps it for ≥ 5 s

3. EXPERIMENTAL SETUP

Baselines (retrained, same split · labels · params)

MODEL	INPUT	PARAMS
Ours	**7 signals**	**114 K**
DRCNN '19	12 PSG ch.	934 K
SelANet '23	7 signals	156 K
Transformer '23	7 signals	149 K
SelANet '23	EDR + SaO2	154 K

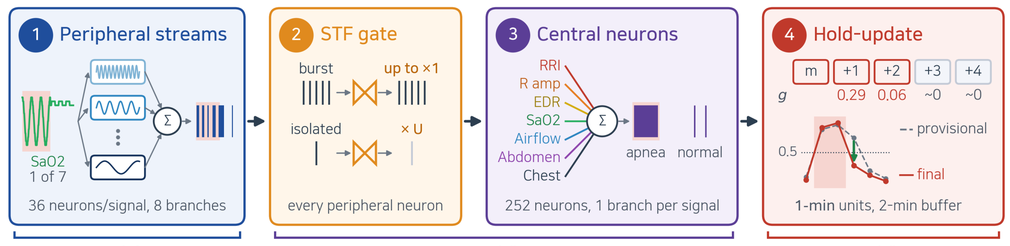
Protocol

- **Subject-level split 694 / 99 / 201, seed 0**
- **Epoch selected on validation**

Metrics

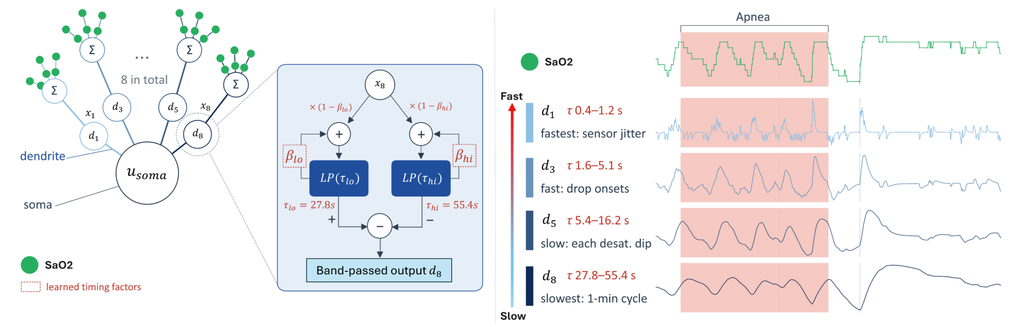
- **ACC · F1 · AUROC · AUPRC at natural prevalence**
- **Subject-level paired bootstrap for difference**

4. PROPOSED ARCHITECTURE

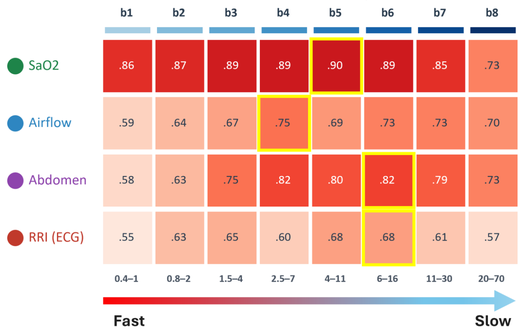


◆4.1 ◆4.2 ◆4.3

◆ 4.1 Peripheral streams – each signal on it owns time scale



Each peripheral neuron splits one signal into 8 dendritic branches, each a learned band-pass filter tuned from fast to slow time scales, and its soma sums them into spikes



◇ Which branch tells apnea?

Each cell is how well one branch's activity separates Apnea from normal breathing.

① Fast branches miss it

Each cell is how well one branch's activity separates apnea from normal breathing (AUROC, test set).

② Slower bands carry it

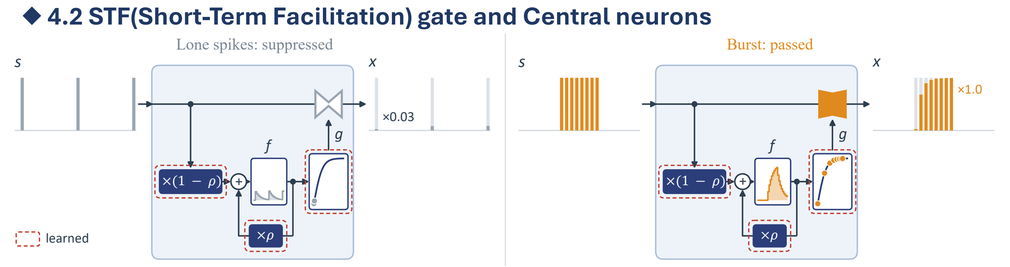
Separation rises to 0.75~0.90 in the mid-to-slow bands

③ Each signal has its own best band

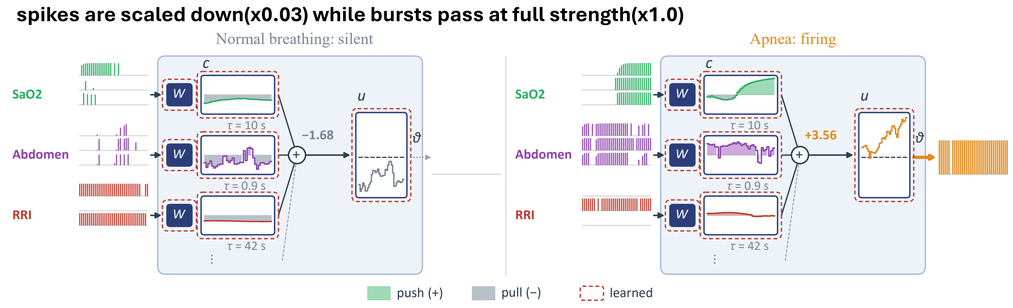
Airflows b4, SaO2 b5, Abdomen and RRI b6

→ Multi-time-scale branches let each signal be read at time scale where apnea appears.

◆ 4.2 STF(Short-Term Facilitation) gate and Central neurons

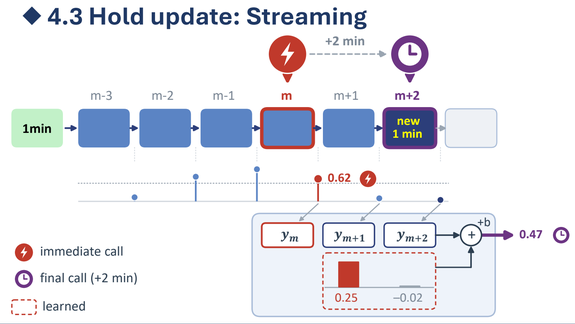


Each peripheral spike passes an STF gate: a facilitation trace builds up only during bursts, so isolated spikes are scaled down(x0.03) while bursts pass at full strength(x1.0)



Signals push (+) or pull (-) on separate branches; the neuron fires when their sum crosses θ

◆ 4.3 Hold update: Streaming



◇ Why hold the call?

Each 1-min unit is called at once, then corrected with the next 2 min

① Streaming, no re-computation

Neuron states carry over, then corrected with the next 2 min

② Immediate call

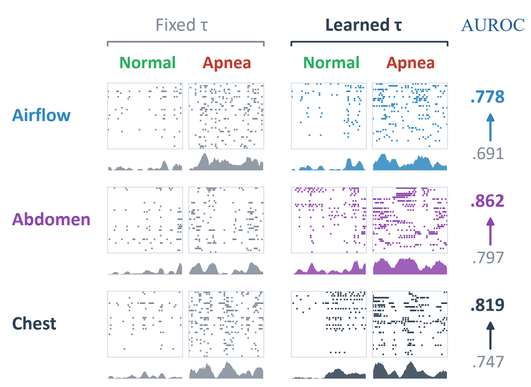
At the end of unit m, the read-out y_m gives an instant alarm

③ Final call (+2 min)

Read-outs of the next 4 units refine the call with learned weights (mostly the next 1min, g=0.29)

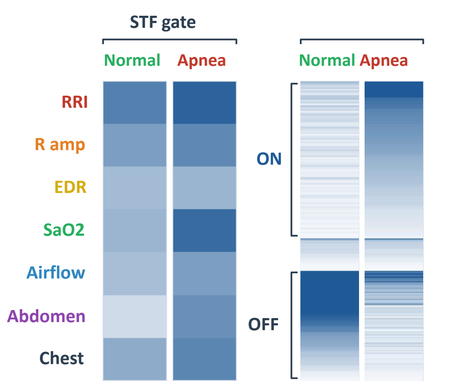
5. RESULTS: MECHANISMS

◆ 5.1 Matched τ makes spikes tell apnea



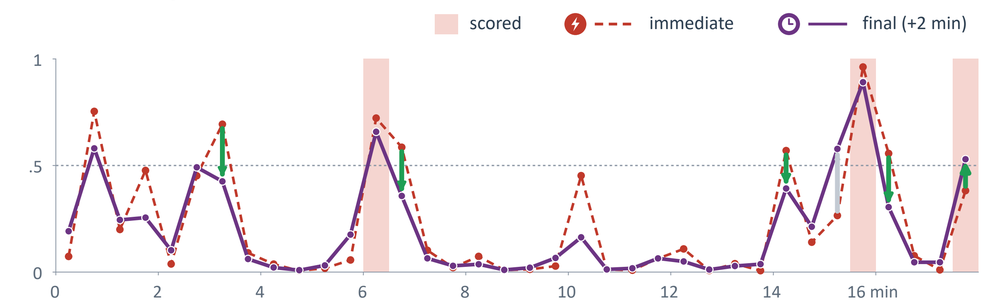
▶ With learned time constants, each signal's spikes separate apnea from normal breathing better

◆ 5.2 Gate opens, central pattern flips



▶ In apnea, the STF gate opens most for SaO2 and abdomen, and central neurons flip from OFF on ON.

◆ 5.3 Hold-update: 2-min call fixes errors

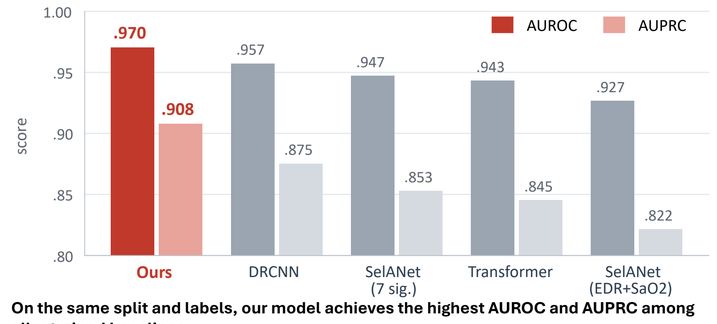


Accuracy %	F1	AUROC	AUPRC
91.8 → **92.2**	.812 → **.820**	.966 → **.970**	.900 → **.908**

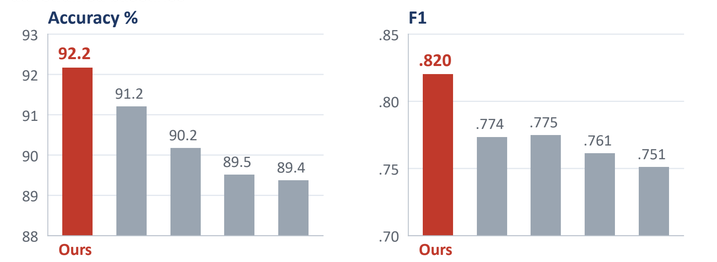
▶ Each 1min unit gets an immediate call, then a final call after the next 2 min of evidence. The final call corrects borderline immediate calls, mostly false alarms, and improves every metric

6. RESULT: PERFORMANCE

◆ 6.1 Best on every metric

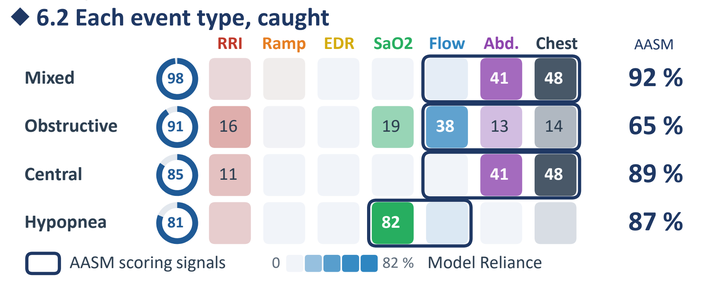


On the same split and labels, our model achieves the highest AUROC and AUPRC among all retrained baselines



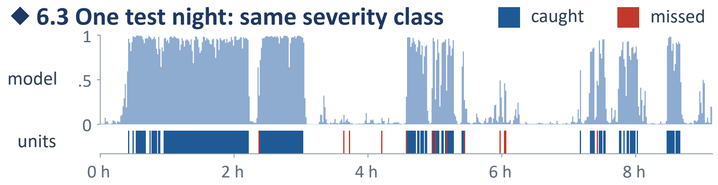
At the 0.5 threshold, it also has the highest accuracy and F1(+0.1%p and + 0.45 over the next best)

◆ 6.2 Each event type, caught



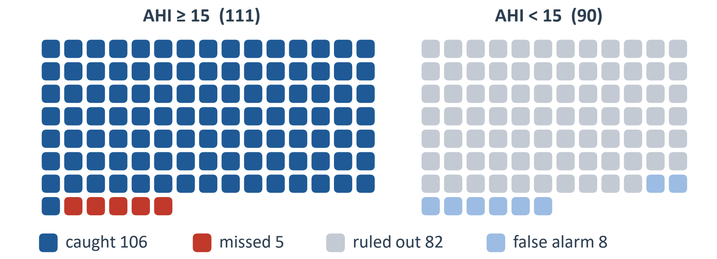
Each event type is caught, and each relies on the signals clinicians use to score it.

◆ 6.3 One test night: same severity class



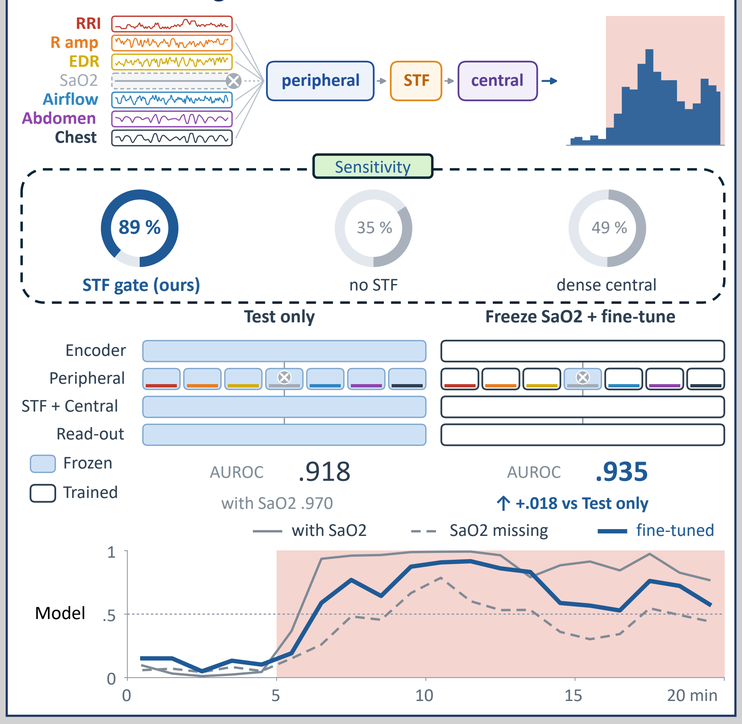
One test night: minute-level calls add up to an estimated AHI of 28.6 vs the scored 28.2, the same moderate severity class

◆ 6.4 Screening moderate-to-severe OSA: 106 of 111 found



Of 111 moderate-to-severe patients, 106 are flagged (sensitivity 95.5%), and 82 of 90 others are correctly ruled out (specificity 91.1%)

◆ 6.5 SaO2 missing: still not blind



7. CONCLUSIONS

1. **Per-signal time scales make peripheral spikes more informative for every signal**
2. **STF gate + per-signal branches let each apnea type be read from the signals clinicians use, and a 2-min hold-update improves every metric**
3. **Best detection among retrained baselines(AUROC 0.970) at 1/8 of DRCNN's size AHI ≥ 15 screening at 95.5% sensitivity, 91.1% specificity.**
4. **Not blind without SaO2: the STF gate keeps 89% of its sensitivity (vs 35% without it), and freeze fine-tuning raises AUROC from .918 to .935.**

8. FUTURE WORK

SHHS	MESA	MrOS
8,444 nights	≈ 2,000 nights	3,933 nights

External validation (NSRR, approved 2026-10) · event-level & apnea-type output · low-power hardware

9. ACKNOWLEDGEMENT

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